SHORT NOTES FOR DAMPED & FORCED OSCILLATION

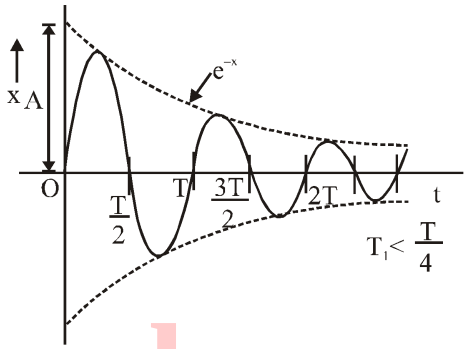
(i) $\frac{d^2x}{dt^2}+2\gamma\frac{dx}{dt}+\omega_0^2x=0$ or $\frac{d^2x}{dt^2}+b\frac{dx}{dt}+\omega_0^2x=0$

$2\gamma=\alpha/m>0$ and ω_0^2, as before is equal to k/m. $\frac{b}{2m}$ or γ is known as damping coefficient

where damping force is –mbv.

(ii) Solution for above equation $x=Ae^{-\gamma t}\sin(\Omega t+\phi)$, Ω is given by $\Omega^2=\omega_0^2-\gamma^2$

(iii)



T_1 is time taken for the particle to reach from mean to extreme position.

(iv) Power Loss in a Damped Oscillation-Q-factor

$$Q=2\pi\times\frac{\text{Average energy stored in one period}}{\text{average loss of energy in that period}}$$

$$Q=\frac{\omega_0}{2\gamma}$$

It may be noted that Q is a dimensionless quantity, greater than unity by our assumption that $\gamma<<\omega_0$.

(v) Time to reach extreme

The natural logarithm of two successive maxima is called logarithmic decrement and is denoted by λ. From Equation it follows that

$$\lambda=\frac{2\pi\gamma}{\Omega}=\frac{\pi}{Q}$$

(vi) Overdamped Motion

In equation if $\gamma>\omega_0$ we do not get oscillatory solution

Critical Damping ($\gamma=\omega_0$)

The solution for the displacement is therefore

$$x=(A+Bt)e^{-\gamma t}$$

It may be seen that once again there are no oscillation. As $t\to\infty$, $x\to 0$ so that the particle takes an infinite time before reaching the equilibrium position.

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(vii) Forced Damped Oscillations

$$\frac{d^2x}{dt^2}+2\gamma\frac{dx}{dt}+\omega_0^2x=\frac{F_0}{m}\cos\omega t$$

$x = C\cos(\omega t + \delta)$

$$C=\frac{F_0}{m}\sqrt{\frac{1}{\left(\omega_0^2-\omega^2\right)^2+4\omega^2\gamma^2}}$$

and $\tan\delta=\frac{2\omega\gamma}{\omega^2-\omega_0^2}$

for maximum amplitude in forced osillation

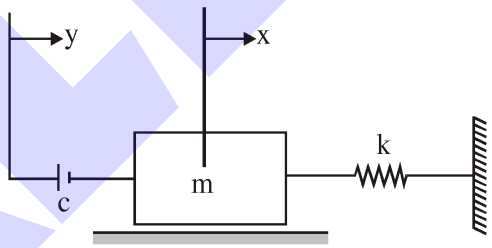
$$\omega=\sqrt{\omega_0^2-2\gamma^2}$$

$$C\approx\frac{F_0}{2m\omega\gamma}=\frac{F_0}{b\omega}\text{ (for weak damping)}$$

SECTION-I

Single Correct Answer Type **15 Q. [3 M (–1)]**

1. The differential equation governing the vibrating system is shown in diagram where c is a viscous damper with damping coefficient c. Select the correct equation of motion.



(A) $m\ddot{x}+c\dot{x}+k(x-y)=0$

(B) $m(\ddot{x}-\ddot{y})+c(\dot{x}-\dot{y})+kx=0$

(C) $m\ddot{x}+c(\dot{x}-\dot{y})+kx=0$

(D) $m(\ddot{x}-\ddot{y})+c(\dot{x}-\dot{y})k(x-y)=0$

2. In a forced vibration with viscous damping, maximum amplitude occurs when forced frequency is:

(A) equal to natural frequency.

(B) slightly less than natural frequency.

(C) slightly greater than natural frequency.

(D) zero

3. A vibrating system consists of a mass 12.5 kg, a spring with stiffness 1000 N/m and a dashpot with damping factor of 15 N-s/m. The value of critical damping of the system is:

(A) 0.223 Ns/m

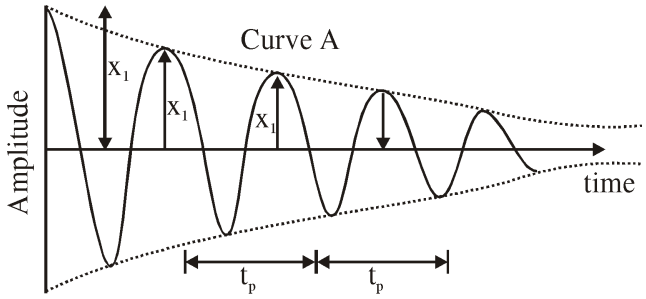
(B) 17.88 Ns/m

(C) 71.4 Ns/m

(D)223.6 Ns/m

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4. The amplitude versus time curve of as damped-free vibration is shown in the below figure. Curve labelled 'A' is :



(A) a longarithmic decrement curve.
(B) an exponentially decreasing curve.
(C) a hyperbolic curve.
(D) a linear curve.

5. When the mass of a critically damped single degree of freedom system is deflected from its equilibrium position and released, it will

(A) return to equilibrium position without oscillation.
(B) oscillate with increasing time period.
(C) oscillate with decreasing amplitude.
(D) oscillate with constant amplitude.

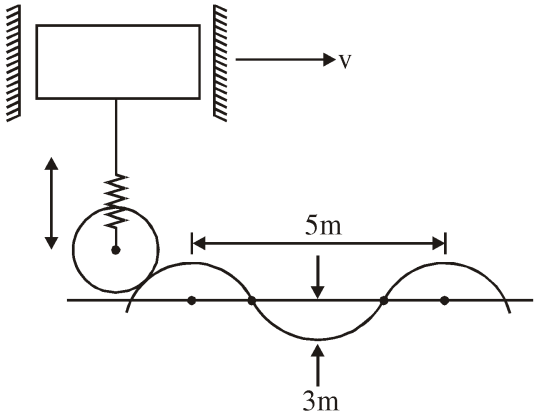
6. If a spring-mass-dashpot system is subjected to excitation by a constant harmonic force, then at resonance, its amplitude of vibration will be:

(A) infinity
(B) inversely proportion to damping
(C) directly proportional to damping
(D) decreasing exponentially with time.

7. In a spring-mass system, the mass 0.1 kg and the stiffness of the spring is 1 kN/m. By introducing a damper, the frquency of oscillation is found to be 90% of the original value. What is the damping factor of the damper?

(A) 1.2 N.s/m (B) 3.4 N.s/m (C) 8.7 N.s/m (D) 12.0 N.s/m

8. The figure below shows the schematic of an automobile having a mass of 900 kg and the suspension spring constant of 81×10^4 N/m. If it travels at a speed of 72 km/hr on a rough road with periodic waviness as shown, what is the forcing frequency of the road on the wheel?



(A) 10 Hz (B) 4 Hz (C) 1.5 Hz (D) 20 Hz

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9. An LCR circuit having a resistor of 20Ω and inductive reactance is 200 Ω. After how many oscillations the ratio of current to its maximum value will become $\frac{1}{e}$?

(A) 10 (B) $\frac{10}{\pi}$

(C) 100 (D) $\frac{100}{\pi}$

10. A spring mass system executes damped harmonic oscillations given by the equation $y = Ae^{-\frac{bt}{2m}} \sin(\omega' t + \phi)$ where the symbols have their usual meanings.

If a 2 kg mass (m) is attached to a spring of force constant (K) 1250 N/m, the period of the oscillation is (π/12)s. The damping constant 'b' has the value.

(A) 9.8 kg/s (B) 2.8 kg/s

(C) 98 kg/s (D) 28 kg/s

11. A simple pendulum of length 1m (consider $\pi^2 = g$) has a time period T when there is no air resistance. If a small air resistance is present during oscillation than :-

(A) The time period will be initially more than 2s and decreases with time.

(B) The time period will be less than 4s initially and increases with time

(C) The time period will be greater than 4s and decrease with time

(D) The time period will be more than 2s and remains constant.

12. A small ball connected to a thread is submerged in a liquid and made to oscillate like a simple pendulum. The time period is T_1 if we consider buoyant force as well as viscous force . If there is no viscous force but there is a buoyant force, the time period is T_2 :

(A) T_1 can be equal to T_2

(B) T_1 is always equal to T_2

(C) $T_1 > T_2$

(D) $T_1 < T_2$

13. Critical damping is the

(A) largest amount of damping for which no oscillation occurs in free vibration

(B) smallest amount of damping for which no oscillation occurs in free vibration

(C) largest amount of damping for which the motion is simple harmonic in free vibration.

(D) smallest amount of damping for which the motion is simple harmonic in free vibration

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14. Which of the following statements are TRUE for damped vibrations?

(P) For a system having critical damping, the system does not undergo a vibratory motion.

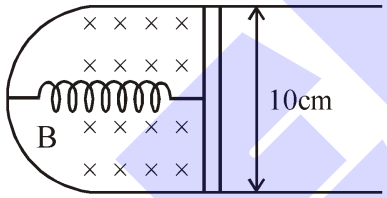
(Q) Logarithmic decrement method is used to determine the amount do damping in a physical system.

(R) In case of damping due to dry friction between moving surfaces resisting force of constant magnitude acts opposite to the relative motion.

(S) For the case of viscous damping, drag force is directly proportional to the square of relative velocity.

(A) P and Q only
(B) P and S only
(C) P, Q and R only
(D) Q and S only

15. A thin strip 10 cm long is on a U shaped wire of negligible resistance and it is connected to a spring of spring constant 0.5 Nm^{-1} (see figure). The assembly is kept in a uniform magnetic field of 0.1 T. If the strip is pulled from its equilibrium position and released, the number of oscillation it performs before its amplitude decreases by a factor of e is N. If the mass of the strip is 50 grams, its resistance 10Ω and air drag negligible, N will be close to : **[JEE Main-2019 April]**



(A) 50000
(B) 5000
(C) 10000
(D) 1000

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Multiple Correct Answer Type **2 Q. [4 M (–1)]**

16. A Harmonic oscillator is represented by the equation $m\frac{d^2x}{dt^2}+\gamma\frac{dx}{dt}+kx=0$ With m = 0.25 kg, γ = 0.070 kg/s and k = 85 N/m. For the given harmonic oscillator.

(A) The time period of oscillation is nearly 0.34 sec.

(B) The number of oscillations in which its mechanical energy will drop to one-half of its initial value is approxmately 7.

(C) The number of oscillations in which its mechanical energy will drop to one-half of its initial value is approxmately 14.

(D) Quality factor of the oscillator is close to 66.

17. A block of mass 1 kg is connected to a horizontal spring of spring constant 100 N/m. The system is made to oscillate in the horizontal direction by oscillating the support point of the spring with an angular frequency of 20 rad/sec and an amplitude of 10 cm. (There is negligible damping)

(A) The amplitude of oscillation of the mass is 3.33 cm

(B) The amplitude of oscillation of the mass is 5 cm

(C) The block oscillates in the same phase as the support point.

(D) The block oscillates with a phase difference of 180° with the support point.

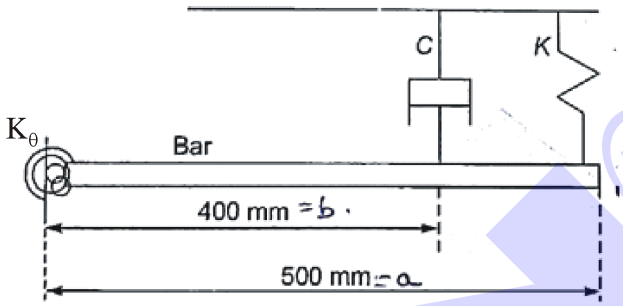
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Linked Comprehension Type (1 Para × 2Q.) [3 M (-1)]

(Single Correct Answer Type)

Paragraph for Q. No. 18 and 19

A uniform rigid slender bar of mass 10 kg, hinged at the left end is suspended with the help of spring and damper arrangement as shown in the figure where K = 2 kN/m, C = 500 Ns/m and the stiffness of the torsional spring K_e is 1 kN /m/rad. Ignore the hinge dimensions.



18. The undamped natural frequency of oscillators of the bar about the hinge point is

(A) 42.43 rad/s (B) 30 rad/s (C) 17.32 rad/s (D) 14.14 rad/s

19. The damping coefficient in the vibration equation is given by

(A) 500 Nms/rad (B) 500 N/(m/s) (C) 80 Nms/rad (D) 80 N/(m/s)

SECTION-II

Linked Comprehension Type (1 Para × 2Q.) [3 M (-1)]

(upto second decimal place)

Paragraph for Q. No. 20 and 21

The energy of recoil of a rocket launcher of mass m = 4500 kg is absorbed in a recoil spring. At the end of the recoil, a damping dashpoot is arranged in such a way that the launcher returns to the firing positin without any oscillation (critical damping). The launcher recoils by a distance of 3m with an initial speed of 10 ms^{-1}.

20. Find the recoil's spring constant (in kN/m)

21. Find the dashpot's coefficient of critical damping. (in kNs m^{-1}) [Given kN = 1000 N]

Numerical Answer Type Question 1Q.[3(0)]

(upto second decimal place)

22. In a spring-mass system, the mass is m and the spring constant is k. The critical damping coefficient of the system is 0.1 kg/s. In another spring-mass system, the mass is 2 m and the spring constant is 8 k. The critical damping coefficient (in kg/s) of this system is

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GR # DAMPED & FORCED OSCILLATION — ANSWER KEY

SECTION-I

Single Correct Answer Type — **15 Q. [3 M (–1)]**

1. (C) 2. (B) 3. (D) 4. (B) 5. (A) 6. (B) 7. (C) 8. (B) 9. (B) 10. (D) 11. (D) 12. (C)
13. (B) 14. (C) 15. (B)

Multiple Correct Answer Type — **2 Q. [4 M (–1)]**

16. (A,B,D) 17. (A,D)

Linked Comprehension Type (Single Correct Answer Type) — **(1 Para × 2Q.) [3 M (-1)]**

18. (A) 19. (D)

SECTION-II

Linked Comprehension Type (upto second decimal place) — **(1 Para × 2Q.) [3 M (-1)]**

20. 50.00 21. 30.00

Numerical Answer Type Question (upto second decimal place) — **1Q.[3(0)]**

22. 0.4

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